

Metapunch X: Combing Multidisplay and Exertion Interaction for Watching and Playing E-sports in Multiverse

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Figure 1: We introduce Metapunch X, an e-sports game that integrates exertion interaction in XR with multidisplays spectating.

ABSTRACT

E-sports events have become more popular. In the Olympics Esports Series, ten items were included in the official competition; four were exergames, and only one implemented extended reality (XR). However, there are no games that apply the encountered-type haptic feedback with exertion interaction. Also, these esports games are only broadcast on screen, resulting in a lack of sense of presence in the virtual world. In our work, we introduce Metapunch X, an encountered-type haptic feedback esports game that integrates exertion interaction in XR with multi-screen spectating. The game

is designed as an asymmetric competition, utilizing an XR head-mounted display (HMD) and a substitutional reality robot to create an esports experience with encountered-type haptic feedback. To enhance the audiences' experience, in addition to the third-person perspective broadcast, we provide 360-degree live streaming available on mobile devices and virtual reality (VR) HMD, allowing audiences to immerse themselves in the virtual environment and experience the competition as if they were personally present at the venue and feel like engaging in a multiverse.

CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms**; Virtual Reality.

KEYWORDS

Exertion Interaction, E-sports, Asymmetric Interaction, Substitutional Reality, Multidisplay

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1 INTRODUCTION

XR developers have created immersive applications and games, blending virtual content with reality, including training simulations and exertion games. The fusion of sports technology enhances athletic performance and opens up new possibilities for sports speculating. [Partridge et al. 2005] employed sensors attached to users to collect data to assess athletes' performance and adjust training programs accordingly. [Zwaan and Barakova 2016] engaged in boxing training with drones, offering different feedback based on user movements. Vcoach [Chen et al. 2022] requires users to wear a VR-HMD, displaying real-time strike data within the HMD, and presenting virtual coaching recommendations. KaraKter [Zhang et al. 2018] used VR-HMD and projection for sports training, emphasizing the effective enhancement of immersion with haptic feedback. [Ozone et al. 2020] proposed a movable short sword robot providing encountered-type haptic feedback for kendo practice. Movablebag [Mendez S et al. 2022] combined VR-HMD with a movable substitutional reality robot for boxing training. [Mueller et al. 2011] emphasized the need for players to understand bodily engagement and interaction in designing exertion games. In addition, this work proposes an immersive experience with a substitutional reality robot based on MovableBag+ for the players to engage in an esports game that requires exertion interaction with encountered-type haptic feedback. At the same time, audiences can spectate the game in several ways, including on-site, on the screen of mobile devices, and inside the VR-HMD. This implementation aims to allow players to engage in the immersive experience and spectators to immerse themselves in the venue to experience the dynamic and charisma of esports events.

2 DESIGN AND IMPLEMENTATION

Metapunch X is a boxing exergame that provides an immersive experience to the players and audiences through multi-display. Apart from general esports games that can only be spectated on the broadcasting screen, audiences can view the competition from different perspectives on various devices. The real-time position and motion of the players and virtual avatar opponents are live-streamed to the screen to provide the visual for the audience on-site and remotely. In addition to the basic third-person perspective, there are various viewpoints and angles can be selected to provide extra choices for spectators. To further enhance the immersiveness, audiences can use their mobile devices or VR-HMD to view the 360-degree live stream to feel like watching the event at a specific point inside the virtual world no matter where they are. Other than bringing a better immersive experience to the audiences, players wearing the XR-HMD (Meta Quest Pro) can view the virtual arena boxing ring and interface on top of the view of real world. The substitutional robot, controlled by another player through a tablet,

is replaced by a virtual avatar inside the XR-HMD, and punches landing on the robot would spawn visual effects that give a stronger feel of impact to the player together with encountered-type haptic feedback to improve the immersiveness of the exertion interaction.

3 APPLICATION: METAPUNCH X

We have designed a competitive game to showcase the application of the system in esports (Figure 1). Participating players are paired up for an asynchronous competition, taking the roles of boxer and dodger by turns with a time limit of one minute. As the boxer, players need to hit targets randomly, spawning around the upper body of the virtual character synchronized with the boxing robot to score points. On the other hand, the dodger has to use a tablet to move the boxing robot within the arena, simulating a real boxer's movement and dodging punches from the opponent. The player who scores higher when acting as the boxer will win. From the audience's view, they can view the actions of the boxer and the dodger in the virtual environment and the game status, like the boxer's current scores and punch speeds. Audiences can select different positions and angles in the virtual world to spectate the event, viewing 360 degrees by moving around the mobile devices or looking around with the VR-HMD to feel immersed in the virtual environment.

4 DISCUSSION AND FUTURE WORK

Our research delved into asynchronous interaction between players, substitutional reality robots, and devising methods for engaging esports enthusiasts. To better understand the level of immersion and participation, we intend to conduct user studies that cater to players and audiences, whether on-site or participating remotely.

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